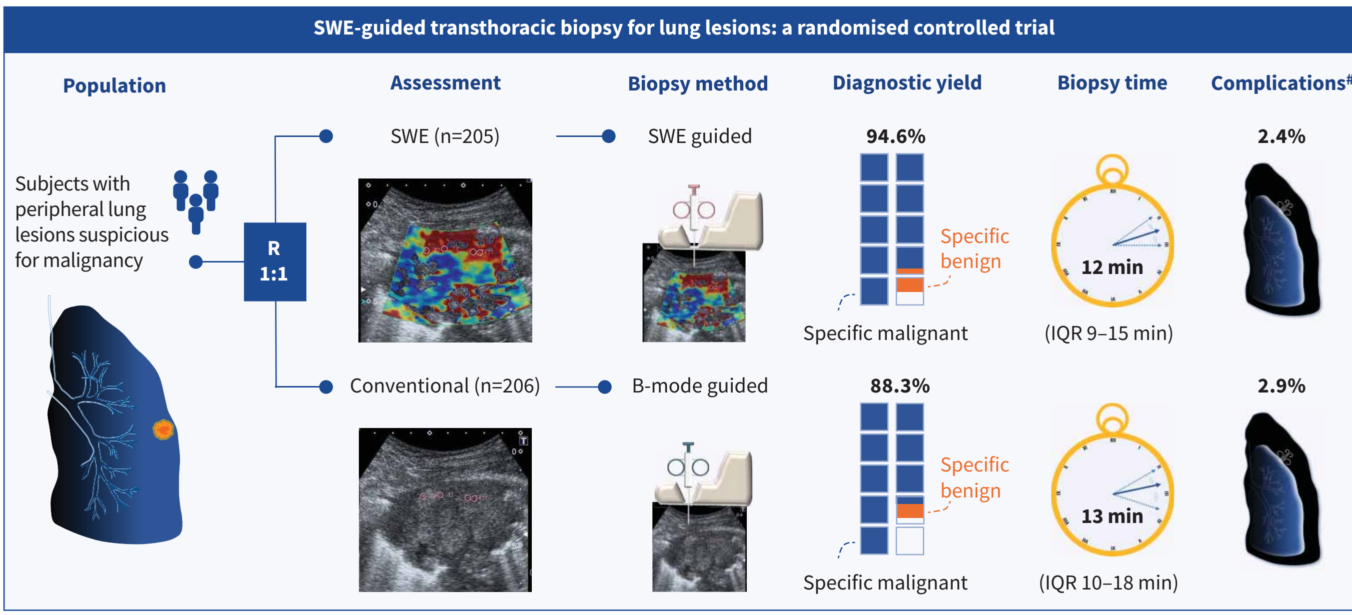
Shear-wave elastography-guided transthoracic biopsy for lung lesions: a randomised controlled trial

Yao-Wen Kuo, Yen-Lin Chen, Yung-Hsuan Chen, Yih-Leong Chang, I-Shiow Jan, Chen-Tu Wu, Min-Shu Hsieh, Chun-Kai Huang, Huey-Dong Wu, Chi-Tai Fang and Hao-Chien Wang



GRAPHICAL ABSTRACT Overview of the study. SWE: shear-wave elastography; R: randomisation; IQR: interquartile range. #: combined incidence of pneumothorax, haemoptysis and haemothorax.

Shear-wave elastography-guided transthoracic biopsy for lung lesions: a randomised controlled trial

Yao-Wen Kuo [1,2], Yen-Lin Chen [3], Yung-Hsuan Chen [3], Yih-Leong Chang[4], I-Shiow Jan[5], Chen-Tu Wu[4], Min-Shu Hsieh[4], Chun-Kai Huang[3], Huey-Dong Wu[1], Chi-Tai Fang[2,6,8] and Hao-Chien Wang[3,7,8]

[1]Department of Integrated Diagnostics and Therapeutics, National Taiwan University Hospital, Taipei, Taiwan. [2]Institute of Epidemiology and Preventive Medicine, College of Public Health, National Taiwan University, Taipei, Taiwan. [3]Department of Internal Medicine, National Taiwan University Hospital, Taipei, Taiwan. [4]Department of Pathology, National Taiwan University Cancer Center, National Taiwan University Hospital and National Taiwan University College of Medicine, Taipei, Taiwan. [5]Department of Laboratory Medicine, National Taiwan University Hospital, Taipei, Taiwan. [6]Division of Infectious Diseases, Department of Internal Medicine, National Taiwan University Hospital, Taipei, Taiwan. [7]Department of Medicine, National Taiwan University Cancer Center, Taipei, Taiwan. [8]Chi-Tai Fang and Hao-Chien Wang contributed equally to this work.

Corresponding author: Hao-Chien Wang (haochienwang@gmail.com)

Check for updates

Shareable abstract (@ERSpublications)
Shear-wave elastography-guided transthoracic biopsy improves diagnostic yield and shortens biopsy duration compared with conventional methods, without increasing complications, offering an efficient diagnostic tool for subpleural lung lesions https://bit.ly/3WHYzcm

Cite this article as: Kuo Y-W, Chen Y-L, Chen Y-H, *et al*. Shear-wave elastography-guided transthoracic biopsy for lung lesions: a randomised controlled trial. *Eur Respir J* 2026; 67: 2402370 [DOI: 10.1183/13993003.02370-2024].

This article has an editorial commentary: https://doi.org/10.1183/13993003.02450-2025

Received: 6 Dec 2024
Accepted: 28 Oct 2025

Abstract

Background Transthoracic shear-wave elastography (SWE) facilitates differentiation between benign and malignant lesions. This pragmatic randomised controlled trial aimed to validate the effectiveness and safety of SWE-guided transthoracic biopsy for diagnosing subpleural lung lesions.

Methods Eligible patients with peripheral lung lesions suspicious for lung malignancy and suitable for biopsy were randomised in a 1:1 ratio to the SWE-guided or conventional ultrasound-guided transthoracic biopsy groups. Rapid on-site cytological evaluations were performed to guide adjustments to biopsy target areas. The primary outcome was the diagnostic yield, defined as the proportion of biopsies yielding a specific benign or malignant diagnosis on cytology or histology. Secondary outcomes included biopsy duration and the combined incidence of post-biopsy complications.

Results From May 2019 to June 2023, a total of 413 patients were randomised. Ultimately, 205 patients in the SWE group and 206 in the conventional group were included in the intention-to-diagnose analysis. The SWE group achieved a significantly higher diagnostic yield compared with the conventional group (94.6% *versus* 88.3%; risk ratio 1.07, 95% CI 1.01–1.14; p=0.02) and a shorter biopsy duration (median (interquartile range) 12 (9–15) *versus* 13 (10–18) min; Hodges–Lehmann median difference −1.0 (95% CI −3.0–0) min; p=0.003). The combined incidence of pneumothorax, haemoptysis and haemothorax was comparable between the groups (2.4% *versus* 2.9%; risk ratio 0.84, 95% CI 0.26–2.70; p=0.77).

Conclusions SWE-guided transthoracic biopsy increased the diagnostic yield and reduced the biopsy duration for subpleural lung lesions without increasing the risk of complications.

Introduction

Ultrasound-guided transthoracic needle biopsy (US-TTNB) has proven to be a reliable and safe diagnostic modality for diagnosing subpleural lung lesions [1–10]. The diagnostic yield of lung US-TTNB is comparable to that of computed tomography (CT)-guided lung biopsy, but with shorter procedure time and lower incidence of post-biopsy pneumothorax [7, 11]. Several thoracic ultrasound modalities have been proposed to aid in differentiating between benign and malignant lung lesions, including colour Doppler ultrasound pulsatile flow signals [12], contrast-enhanced ultrasound (CEUS) [13] and ultrasound elastography [14, 15]. During US-TTNB, colour Doppler is also used to identify intercostal vessels and intralesional vasculature, assisting in preventing vessel puncture [16, 17]. In addition, rapid on-site cytological evaluation (ROSE) facilitates assessment of specimen adequacy during biopsy and may improve diagnostic accuracy [18].

Two-dimensional shear-wave elastography (SWE) has been used to image subpleural lung lesions [14], pleural diseases [19] and tuberculous lymphadenitis [20] in pulmonology. Concerns previously existed regarding the use of elastography for lung lesions, as acoustic radiation force elastography could induce superficial subpleural haemorrhage in animal studies [21, 22]. In contrast, gas-free fetal mouse lungs did not exhibit the shock wave-related damage [23]. Our prior cohort study validated the predictive value of SWE in differentiating malignant from benign subpleural lung lesions, with no increased incidence of haemoptysis or pneumothorax [14].

Heterogeneity within a lung lesion may compromise the diagnostic yield of US-TTNB in daily practice, as biopsies may inadvertently target necrotic regions or areas affected by obstructive pneumonia associated with malignancy. Aiming to validate the effectiveness and safety of SWE-guided transthoracic biopsy in real-world practice, this pragmatic randomised controlled trial compared SWE-guided transthoracic biopsy with conventional US-TTNB for subpleural lung lesions.

Methods

Trial design

This study was designed as a prospective, parallel-group, pragmatic randomised controlled trial with a 1:1 allocation ratio and double-blinding of participants and outcome assessors. This trial is registered on ClinicalTrials.gov (NCT03881410). Eligible patients were enrolled by study investigators at the Chest Ultrasound Unit of National Taiwan University Hospital (NTUH), Taipei, Taiwan. The study protocol was approved by the NTUH Research Ethics Committee A (NTUH-REC 201812180RINA) and all participants provided written informed consent. Reporting conforms to the CONSORT (Consolidated Standards of Reporting Trials) 2010 guidelines (www.consort-spirit.org).

Participants

The inclusion criteria were: 1) chest CT showing a peripheral lung lesion suspicious for lung malignancy; and 2) a lesion that was sonographically accessible and suitable for US-TTNB. Suitability was determined by the operator, considering lesion size, margin clarity and anatomical factors, such as prominent intralesional vessels and proximity to the heart or major vessels. Exclusion criteria were: 1) age <20 years; 2) anticipated poor shear-wave propagation in the presence of cardiac or great vessel pulsation, a thick chest wall, or pleural effusion [14]; and 3) inability to hold breath for 5 s. Enrolment was conducted during staffed hours. Patients were not enrolled when simultaneous procedures exceeded available staff capacity.

Interventions

Pulmonary medicine fellows and early-career pulmonologists, under the supervision of senior pulmonologists, performed B-mode ultrasound, SWE and US-TTNB (see supplementary figure E1 for study protocol). Pre-procedural bleeding risk was assessed by medication review, coagulation profile and colour Doppler mapping of intralesional vessels. Baseline clinical features, including age, gender, body weight and height, were recorded.

B-mode and SWE were performed on an Aplio 500 Platinum ultrasound machine (Toshiba Medical Systems, Tochigi-Ken, Japan) using a PVT-375BT convex transducer (1.5–6.1 MHz; see supplementary table E1 for ultrasound exposure conditions). B-mode ultrasound assessed lesion size, location, chest wall thickness and echogenicity of subpleural lung lesions. In the SWE group, patients were instructed to hold their breath for 5 s while SWE was initiated and adequate shear-wave propagation verified on the arrival time contour map (elastogram). Four 3-mm circular regions of interest (ROIs) were then placed in the visually red-coded areas of higher stiffness on the fixed 0–120 kPa elastogram. The mean elasticity of these ROIs was recorded across three acquisitions and averaged to yield the final stiffness value. Biopsy target areas were identified within red-coded, high-stiffness regions on the elastogram (supplementary figures E2 and E3) [14]. In the conventional group, biopsy target areas were selected under B-mode ultrasound guidance to sample hypoechoic solid components and the mural regions of centrally necrotic lesions.

For US-TTNB, operators switched to a PLT-308BTP linear puncture transducer (2.0–5.5 MHz) (Toshiba), which is used for needle visualisation and does not support SWE. Biopsies were obtained under real-time ultrasound guidance with an 18-gauge SuperCore semi-automatic biopsy needle (Argon Medical Devices, Plano, TX, USA) with an adjustable specimen notch (cutting) length of 10 or 20 mm. ROSE was performed by senior pulmonologists on imprint smears from the first two needle passes, with smear results (benign, malignant or inconclusive) guiding adjustments to biopsy target areas. The first ROSE guided the second pass, and the second ROSE guided subsequent passes. A minimum of six passes was required to secure tissue for next-generation sequencing, and all imprint smears and core specimens were finally reviewed by cytopathologists and histopathologists.

Additional details regarding the intervention and the education programme are provided in the supplementary material.

Outcomes

The primary outcome was diagnostic yield, defined as the proportion of biopsies yielding a specific pathological diagnosis (malignant or benign) on cytology or histology [24], calculated as (specific malignant+specific benign)/total patients. Both intention-to-diagnose and per-protocol analyses were conducted. Prespecified subgroup analyses compared diagnostic yield by final diagnosis (benign *versus* malignant) and by lesion size (⩽3 *versus* >3 cm). Secondary outcomes included biopsy duration, defined as the time from skin disinfection to biopsy completion, and the combined incidence of pneumothorax, haemoptysis and haemothorax.

Specific lung malignancies included adenocarcinoma, small cell carcinoma, squamous cell carcinoma, non-small cell lung carcinoma not otherwise specified, lung metastases, lymphoma and other malignancies. Specific benign lesions included identifiable microorganisms (*e.g. Cryptococcus*, bacteria, fungi or acid-fast bacilli), granulomatous inflammation, organising pneumonia or specific benign tumours. Non-diagnostic results included atypia, minimal histological changes, suboptimal specimens, necrosis, inflammatory change, necrotising inflammation, intra-alveolar fibrinous exudate, diffuse alveolar damage, coagulation necrosis and none made.

Final diagnoses were established through cytological or histological examination per various diagnostic modalities, microbiological studies or assessment of the patient's clinical course over a 6-month follow-up after US-TTNB.

Sample size

Sample size calculations assumed an 80% power (two-sided α=0.05) to detect a difference in diagnostic yield between conventional (75%) and SWE-guided (90%) biopsy. Initially, we planned to enrol 200 patients over 2 years. After trial initiation, updated variance estimates from the literature and our prior work prompted a reassessment, revising the assumed yields to 80% for the conventional group and 90% for the SWE group. Consequently, the target sample size was increased to 398 participants while retaining the original α and β thresholds. For operational planning, we prespecified an enrolment range of ±10% around this target (*i.e.* 358–438). The recruitment period was extended to 3 years, and no interim analyses were performed.

Randomisation

The randomisation table was created using the SAS PROC PLAN (SAS, Cary, NC, USA) with permuted blocks of six and stratified by gender. Participants were then allocated 1:1 to the SWE group or conventional group *via* the REDCap randomisation module, which also maintained allocation concealment from the enrolling investigators. Sequence generation and the implementation of the concealment mechanism were managed by the Clinical Trial Center of NTUH.

Blinding

The physician performing the biopsy also conducted B-mode ultrasound and elastography for patients allocated to the SWE group, and therefore could not be blinded to allocation. To minimise detection bias, both patients and outcome assessors (cytopathologists and histopathologists) remained blinded to group assignment.

Statistical methods

Categorical variables are presented as number (percentage) and compared using Pearson's Chi-squared or Fisher's exact test. Continuous variables are summarised as mean with standard deviation or median with interquartile range (IQR) and compared by the t-test or Wilcoxon–Mann–Whitney test, respectively. The diagnostic yield and the incidence of complications were analysed with the Chi-squared or Fisher's exact test, and biopsy duration was analysed with the Wilcoxon–Mann–Whitney test. Effect sizes for binary outcomes are expressed as risk ratios and risk differences with 95% confidence intervals; biopsy duration is summarised by the Hodges–Lehmann estimate of median difference with its 95% confidence interval. Stata version 16.1 (StataCorp, College Station, TX, USA) was used for analysis, with a two-sided p-value ⩽0.05 considered statistically significant.

Results

Baseline characteristics of the patients and lung lesions

Patient recruitment occurred from May 2019 to June 2023, with follow-up completed in December 2023 (figure 1). The trial was initially planned for a 3-year duration. However, the study timeline was extended by 1 year due to the COVID-19 pandemic. Of the 935 patients with chest CT findings suspicious for lung malignancies who were screened, 413 patients were enrolled in the study (see supplementary table E2 for the subset of excluded patients not shown in the CONSORT diagram). Final diagnoses could not be established for two patients due to loss to follow-up. A total of 411 patients were analysed (205 in the SWE group and 206 in the conventional group). The mean±SD age was 66.1±12.9 years and the mean±SD body mass index was 22.6±4.5 $kg \cdot m^{-2}$. Most patients (63.7%) were men. Most lung lesions were hypoechoic (92.5%) on B-mode images. The mean±SD width and vertical dimension of the lung lesions were 57.2±22.2 and 53.5±21.8 mm, respectively. The three most common lesion locations were the right upper lobe (33.3%), left upper lobe (22.9%) and right lower lobe (21.7%). Baseline clinical and laboratory characteristics were comparable between the two groups (table 1). The mean±SD elasticity in the SWE group was 106.2±27.9 kPa.

Primary outcome: diagnostic yield

This pragmatic study involved 12 pulmonologists and 24 pulmonary medicine fellows as operators. Operator distribution was comparable between groups. Board-certified pulmonologists served as operators in 115 SWE and 118 conventional cases (p=0.89). In the intention-to-diagnose analysis, a specific diagnosis (table 2) was achieved in 194 patients (94.6%) in the SWE group and 182 patients (88.3%) in

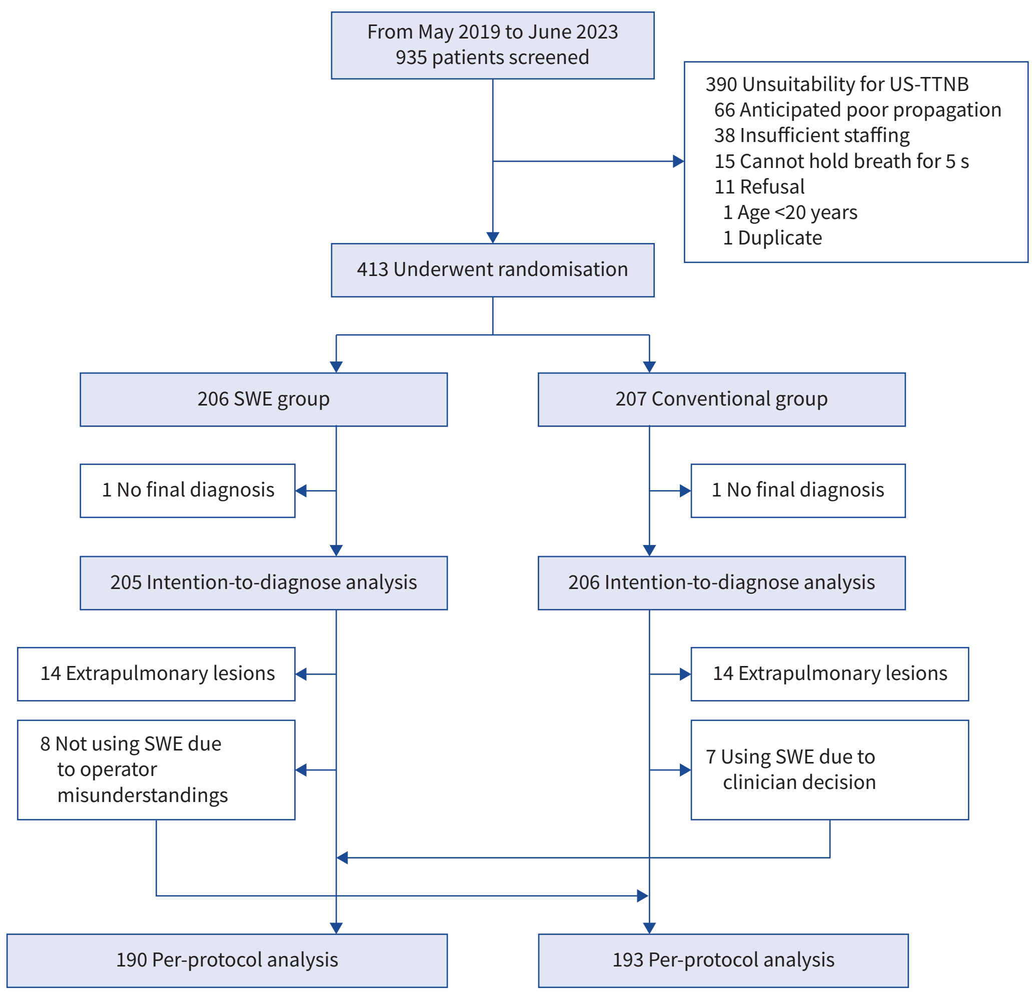


FIGURE 1 Flow diagram illustrating patient enrolment in this study. US-TTNB: ultrasound-guided transthoracic needle biopsy; SWE: shear-wave elastography.

TABLE 1 Baseline characteristics of the patients and lung lesions

	SWE group (n=205)	Conventional group (n=206)
Age, years	65.3±13.4	66.8±12.3
Male	136 (66.3)	126 (61.2)
Body mass index, kg·m^{-2}	22.3±4.0	23.0±4.9
Chest wall thickness, mm	19.9±7.4	21.0±7.3
Lesions on B-mode		
Hypoechoic	190 (92.7)	190 (92.2)
Air bronchogram	15 (7.3)	16 (7.8)
Width, mm	59.0±23.2	55.4±21.1
Vertical dimension, mm	53.2±23.4	53.8±20.2
Elasticity, kPa	106.2±27.9	
Location[#]		
Right upper lobe	79 (38.5)	58 (28.2)
Right middle lobe	7 (3.4)	13 (6.3)
Right lower lobe	41 (20.0)	48 (23.3)
Left upper lobe	48 (23.4)	46 (22.3)
Left lower lobe	30 (14.6)	41 (19.9)
Laboratory		
Platelets, ×10^3 μL^{-1}	314.3±132.9	315.2±129.8
Prothrombin time, s	10.9±0.7	10.9±0.6
Partial thromboplastin time, s	29.2±2.6	29.1±2.7
INR	1.0±0.1	1.0±0.1
Blood urea nitrogen, mg·dL^{-1}	17.8±11.0	16.6±8.5
Creatinine, mg·dL^{-1}	1.0±1.0	0.9±0.7

Data are presented as mean±SD or n (%). SWE: shear-wave elastography; INR: International Normalised Ratio. [#]: 14 biopsied lesions in the SWE group and 14 in the conventional group were ultimately determined to be extrapulmonary.

the conventional group (risk ratio 1.07, 95% CI 1.01–1.14; risk difference 6.3%, 95% CI 0.9–11.6%; p=0.02). Among patients whose initial US-TTNB yielded a specific benign result, 1/15 (6.7%) in the SWE group and 1/12 (8.3%) in the conventional group were ultimately malignant and were therefore excluded from the numerator of diagnostic yield. Among patients whose initial US-TTNB yielded a non-diagnostic result, 6/10 (60.0%) in the SWE group and 18/23 (78.3%) in the conventional group were ultimately malignant. Methods used to establish the final diagnosis are summarised in supplementary table E3.

In the prespecified subgroup analyses of final malignant diagnoses, diagnostic yield was 96.3% (180/187) in the SWE group and 90.0% (171/190) in the conventional group (risk ratio 1.07, 95% CI 1.01–1.13; risk difference 6.3%, 95% CI 1.2–11.3%; p=0.02). Adenocarcinoma and squamous cell carcinoma were the most frequent malignant diagnoses. Among final benign diagnoses, diagnostic yield was 77.8% (14/18) in the SWE group and 68.8% (11/16) in the conventional group (risk ratio 1.13, 95% CI 0.75–1.71; risk difference 9.0%, 95% CI −20.7–38.8%; p=0.55). Common specific benign diagnoses were granulomatous inflammation, organising pneumonia and fungal infection. For lesions >3 cm, diagnostic yield was 95.5% (169/177) in the SWE group and 89.6% (165/183) in the conventional group (risk ratio 1.07, 95% CI 1.00–1.13; risk difference 5.9%, 95% CI 0.5–11.2%; p=0.03). For lesions ⩽3 cm, diagnostic yield was 89.3% (25/28) in the SWE group and 78.3% (18/23) in the conventional group (risk ratio 1.14, 95% CI 0.89–1.47; risk difference 11.0%, 95% CI −9.4–31.4%; p=0.28).

Among the biopsied lesions, 28 were ultimately determined to be extrapulmonary lesions, comprising pleural (SWE n=4, conventional n=5), mediastinal (SWE n=3, conventional n=6) and chest wall (SWE n=7, conventional n=3) subtypes. Protocol deviations were identified in 15 patients: eight in the SWE group did not use SWE guidance due to operator misunderstandings, while seven in the conventional group underwent SWE-guided biopsy due to clinician decisions (figure 1). The per-protocol analysis excluded extrapulmonary lesions and included 190 patients in the SWE group and 193 in the conventional group (supplementary table E4). Diagnostic yield was 94.7% (180/190) in the SWE group and 88.1% (170/193) in the conventional group (risk ratio 1.08, 95% CI 1.01–1.14; risk difference 6.7%, 95% CI 1.1–12.2%; p=0.02). In the SWE group, the mean±SD elasticity of malignant lesions was 107.4±27.4 kPa (n=188) compared to 89.9±33.3 kPa (n=17) for benign lesions.

TABLE 2 Diagnostic yield (intention-to-diagnose analysis) and final diagnoses

	SWE group (n=205)	Conventional group (n=206)	Effect size (95% CI)	p-value
Primary outcome				
Diagnostic yield[#]	194/205 (94.6)	182/206 (88.3)	RR 1.07 (1.01–1.14) RD 6.3% (0.9–11.6%)	0.02
Specific malignant (initial US-TTNB)	180/205 (87.8)	171/206 (83.0)		
Specific benign (initial US-TTNB)	15/205 (7.3)	12/206 (5.8)		
Non-diagnostic (initial US-TTNB)[¶]	10/205 (4.9)	23/206 (11.2)		
Subgroup analyses (diagnostic yield by final diagnosis and size)[+]				
Final malignant	180/187 (96.3)	171/190 (90.0)	RR 1.07 (1.01–1.13) RD 6.3% (1.2–11.3%)	0.02
Final benign	14/18 (77.8)	11/16 (68.8)	RR 1.13 (0.75–1.71) RD 9.0% (−20.7–38.8%)	0.55
Lesions >3 cm	169/177 (95.5)	164/183 (89.6)	RR 1.07 (1.00–1.13) RD 5.9% (0.5–11.2%)	0.03
Lesions ⩽3 cm	25/28 (89.3)	18/23 (78.3)	RR 1.14 (0.89–1.47) RD 11.0% (−9.4–31.4%)	0.28
Final diagnoses				
Malignant				0.38
Adenocarcinoma	79/187 (42.2)	78/190 (41.1)		
Small cell	9/187 (4.8)	15/190 (7.9)		
Squamous cell	41/187 (21.9)	33/190 (17.4)		
NSCLC-NOS	6/187 (3.2)	9/190 (4.7)		
Metastatic	16/187 (8.6)	10/190 (5.3)		
Lymphoma	7/187 (3.7)	5/190 (2.6)		
Others[§]	29/187 (15.5)	40/190 (21.1)		
Specific benign				0.25
Granulomatous inflammation	1/14 (7.1)	5/11 (45.5)		
Organising pneumonia	4/14 (28.6)	2/11 (18.2)		
Benign tumours	2/14 (14.3)	1/11 (9.1)		
Fungal	3/14 (21.4)	3/11 (27.3)		
Acid-fast bacilli	1/14 (7.1)	0		
Bacteria	3/14 (21.4)	0		

Data are presented as n/n (%), unless otherwise stated. SWE: shear-wave elastography; RR: risk ratio; RD: risk difference; US-TTNB: ultrasound-guided transthoracic needle biopsy; NSCLC-NOS: non-small cell lung carcinoma not otherwise specified. [#]: proportion of patients whose first US-TTNB gave a specific diagnosis (either malignant or benign). [¶]: non-diagnostic results included atypia (2 *versus* 2), minimal histological change (0 *versus* 4), suboptimal specimen (0 *versus* 1), necrosis (2 *versus* 1), inflammatory change (3 *versus* 6), necrotising inflammation (0 *versus* 1), fibrinous exudate (2 *versus* 4), diffuse alveolar damage (1 *versus* 2), coagulation necrosis (0 *versus* 1) and none made (0 *versus* 1), totalling 10 in the SWE group and 23 in the conventional group. [+]: final diagnoses were established by a composite reference standard (*via* various diagnostic modalities, positive microbiology or ⩾6-month clinico-radiological follow-up after US-TTNB). In the final malignant and benign rows, the denominator is all patients whose final diagnosis was malignant or benign, respectively; the numerator is those whose first US-TTNB already yielded the matching specific diagnosis. [§]: other malignancies included pleomorphic carcinoma, lymphoepithelioma-like carcinoma, sarcomatoid carcinoma, poorly differentiated carcinoma, neuroendocrine carcinoma, spindle cell malignancy, invasive mucinous adenocarcinoma, lung adenocarcinoma combined with small cell carcinoma, adenosquamous cell carcinoma, malignant mesothelioma, malignant solitary fibrous tumour and thymoma.

Secondary outcome: biopsy duration and related procedural parameters

Median (IQR) biopsy duration (table 3) was 12 (9–15) min in the SWE group and 13 (10–18) min in the conventional group (Hodges–Lehmann median difference −1.0 (95% CI −3.0–0) min); p=0.003). Biopsy target changes based on ROSE of the first biopsy pass were observed in 9.8% of patients in the SWE group and 32.7% of patients in the conventional group ($p<0.001$). The total number of biopsy passes was similar between the SWE group and the conventional group (6.2±0.1 *versus* 6.1±0.1 passes; p=0.73) (see supplementary table E5 for the diagnostic performance of the first biopsy pass and the second biopsy pass combined with subsequent biopsy passes).

Secondary outcome: combined incidence of complications

The incidence of complications between the two groups is presented in table 4. Post-biopsy pneumothorax developed in three patients (1.46%) in the SWE group and four patients (1.94%) in the conventional group, all resolving without drainage. Post-biopsy haemoptysis occurred in one patient (0.49%) in the

TABLE 3 Secondary outcome: biopsy duration and related procedural parameters

	SWE group (n=205)	Conventional group (n=206)	Effect size (95% CI)	p-value
Biopsy duration, min	12 (9–15)	13 (10–18)	Hodges–Lehmann median difference −1.0 (−3.0–0)	0.003
Biopsy target change	20 (9.8)	67 (32.5)		<0.001
Total biopsy passes	6.2±0.1	6.1±0.1		0.73

Data are presented as median (interquartile range), n (%) or mean±SD, unless otherwise stated. SWE: shear-wave elastography.

SWE group and two patients (0.97%) in the conventional group. Of these three patients, one in the conventional group received inhaled tranexamic acid, while haemoptysis resolved spontaneously in the other two. One patient (0.52%) with chronic kidney disease in the SWE group developed post-biopsy haemothorax requiring chest tube insertion, whereas no patients were reported in the conventional group. The combined incidence of pneumothorax, haemoptysis and haemothorax was not significantly different between the two groups (2.4% in the SWE group *versus* 2.9% in the conventional group; risk ratio 0.84, 95% CI 0.26–2.70; risk difference −0.5%, 95% CI −3.6–2.6%; p=0.77).

Discussion

This study validated the effectiveness and safety of SWE-guided transthoracic biopsy for subpleural lung lesions, achieving a higher diagnostic yield and shorter biopsy duration compared to conventional US-TTNB, without increasing the risk of complications.

The findings of this randomised trial are consistent with our prior cohort study [14]. Targeted lung lesions in this study were clearly identifiable under B-mode ultrasound. The combined incidence of pneumothorax, haemoptysis and haemothorax did not differ significantly between groups. This is the first randomised controlled study to demonstrate the clinical utility and impact of SWE-guided transthoracic biopsy for subpleural lung lesions in real-world practice, improving diagnostic yield and procedural efficiency without increasing the complication rate.

In this study, as ROSE was systematically applied in both arms, it may have improved overall diagnostic accuracy but also reduced the observable difference between SWE and conventional ultrasound. The benefit of SWE might be even more pronounced in settings where ROSE is not routinely available. While the total number of biopsy passes was similar between the groups, the SWE group had a significantly lower percentage of biopsy target changes, which may have contributed to the shorter biopsy duration. In addition, the diagnostic yield was significantly higher in the SWE group than in the conventional group, highlighting the added diagnostic value of SWE in evaluating subpleural lung lesions, beyond the role of ROSE. Furthermore, this pragmatic randomised controlled trial conducted in routine practice by 12 pulmonologists and 24 pulmonary medicine fellows suggests that SWE may reduce operator dependency [25].

In the prespecified subgroup analyses, SWE improved diagnostic yield for malignant lesions and showed a non-significant trend for benign lesions, likely due to the limited sample size. The elasticity of benign lesions in this study exceeded the 65 kPa cut-off reported previously [14]. This may be explained by the predominance of organising pneumonia and granulomatous inflammation, which typically exhibit higher elasticity than other benign aetiologies [14]. SWE also improved diagnostic yield for lesions >3 cm and

TABLE 4 Secondary outcome: combined incidence of post-biopsy complications

	SWE group (n=205)	Conventional group (n=206)	Effect size (95% CI)	p-value
Combined	5 (2.4)	6 (2.9)	RR 0.84 (0.26–2.70) RD −0.5% (−3.6–2.6%)	0.77
Pneumothorax	3 (1.5)	4 (1.9)		1.00
Haemoptysis	1 (0.5)	2 (1.0)		1.00
Haemothorax	1 (0.5)	0		0.50

Data are presented as n (%), unless otherwise stated. SWE: shear-wave elastography; RR: risk ratio; RD: risk difference.

showed a non-significant trend for lesions ⩽3 cm. The subgroup analysis for lesions ⩽3 cm is likely underpowered, and generalisability to small or technically challenging lesions remains uncertain. Future studies are needed to evaluate SWE's diagnostic role in benign lesions and in smaller or anatomically challenging lesions. Because SWE maps tissue stiffness and CEUS delineates perfused regions [26, 27], future trials should also integrate both SWE and CEUS. Given their complementary information, this is a promising area for investigation.

The strengths of this study include its randomised design, blinding of participants and outcome assessors, and incorporation of real-world clinical settings with 12 pulmonologists and 24 pulmonary medicine fellows as operators, which enhances the generalisability of its findings. However, this study has four limitations. First, this was a single-centre study in Taiwan, conducted in the Chest Ultrasound Unit (NTUH) with >40 years of US-TTNB experience. Although operator expertise was balanced between groups in this trial, reproducibility in other centres may vary and outcomes may depend on operator experience even under a standardised operating procedure (SOP). Second, anticipated unacceptable shear-wave propagation, such as in lesions within effusion, near great vessels, deep lesions such as central tumours with obstructive pneumonitis, and in patients with thick chest walls [14], were excluded from this study, limiting the broader applicability of the findings. Third, to minimise post-randomisation exclusions [28], 15 protocol deviations and 28 extrapulmonary lesion diagnoses were included in the primary intention-to-diagnose analysis. Nevertheless, the per-protocol analysis also confirmed the superiority of the SWE-guided biopsy group over the conventional group, highlighting the robustness of this study. Fourth, because SWE imaging was performed with a convex probe and biopsy guidance with a linear probe, a minor mismatch between stiffness mapping and the final needle trajectory cannot be entirely excluded. However, this limitation is likely negligible given that most lesions were large and subpleural.

Beyond these methodological limitations, practical considerations regarding cost and implementation merit discussion. As SWE is a software upgrade on existing high-end ultrasound systems, the incremental capital cost is lower than purchasing a full CT-guided biopsy system. Disposable costs remain identical to conventional US-TTNB, making SWE a radiation-free, relatively low-cost option for centres already performing ultrasound-guided procedures. However, implementation may be limited by equipment availability and operator training. In our ultrasound unit, only two of the four ultrasound systems support two-dimensional SWE. Because elastograms are not displayed in real-time, operators must memorise red-coded stiffness zones for subsequent B-mode targeting. Therefore, structured training in breath-hold technique, propagation map verification, ROI placement and safe needle trajectory planning, provided under a formal SOP and supervised practice, is essential.

In conclusion, this pragmatic randomised controlled study demonstrated the superiority of SWE-guided transthoracic biopsy over conventional US-TTNB for diagnosing subpleural lung lesions. The SWE group achieved a higher diagnostic yield and shorter biopsy duration compared with the conventional group with comparable incidence of complications. These findings suggest that SWE-guided transthoracic biopsy can be considered the preferred approach for subpleural lung lesions due to its superior diagnostic performance and procedural efficiency.

Acknowledgements: We sincerely thank Chin-Hao Chang and Yu-Han Chang from the Clinical Trial Center of National Taiwan University Hospital (NTUH), Taipei, Taiwan for generating the randomisation sequence and ensuring allocation concealment using the REDCap system. We appreciate the clinical support provided by the staff at the Chest Ultrasound Unit (NTUH). This work is part of the PhD dissertation of the first author Yao-Wen Kuo (National Taiwan University; 2023).

Data availability: De-identified individual participant data that underlie the results reported in this article, including data dictionaries, will be available upon reasonable request. Additionally, the study protocol, statistical analysis plan, Stata analytic code and informed consent forms will be shared. Data access will begin 3 months after publication and remain available for 5 years. Researchers with methodologically sound proposals may access the data to achieve aims in the approved proposal. Requests for access should be directed to kyw@ntu.edu.tw, and requestors will be required to sign a data access agreement.

This clinical trial is prospectively registered at ClinicalTrials.gov with identifier number NCT03881410.

Author contributions: Y-W. Kuo, C-T. Fang and H-C. Wang contributed to the study design, data acquisition, analysis and interpretation, as well as drafting the manuscript. Y-L. Chen, Y-H. Chen, Y-L. Chang, I-S. Jan, C-T. Wu, M-S. Hsieh, C-K. Huang and H-D. Wu contributed to data acquisition and interpretation. All authors participated in the critical revision of the manuscript for important intellectual content.

Ethics statement: This study was approved by the NTUH Research Ethics Committee A (NTUH-REC 201812180RINA) and conducted in accordance with the Declaration of Helsinki. Written informed consent was obtained from all participants or their legal surrogates.

Conflict of interest: The authors have no potential conflicts of interest to declare.

Support statement: This work was financially supported by the National Science and Technology Council, ROC (108-2314-B-002-193-, 110-2314-B-002-258-MY3). Funding information for this article has been deposited with the Open Funder Registry.

References

1 Yang PC. Ultrasound-guided transthoracic biopsy of the chest. *Radiol Clin North Am* 2000; 38: 323–343.
2 Liao WY, Chen MZ, Chang YL, *et al.* US-guided transthoracic cutting biopsy for peripheral thoracic lesions less than 3 cm in diameter. *Radiology* 2000; 217: 685–691.
3 Diacon AH, Theron J, Schubert P, *et al.* Ultrasound-assisted transthoracic biopsy: fine-needle aspiration or cutting-needle biopsy? *Eur Respir J* 2007; 29: 357–362.
4 Laursen CB, Naur TM, Bodtger U, *et al.* Ultrasound-guided lung biopsy in the hands of respiratory physicians: diagnostic yield and complications in 215 consecutive patients in 3 centers. *J Bronchology Interv Pulmonol* 2016; 23: 220–228.
5 Corcoran JP, Tazi-Mezalek R, Maldonado F, *et al.* State of the art thoracic ultrasound: intervention and therapeutics. *Thorax* 2017; 72: 840–849.
6 Christiansen IS, Clementsen PF, Bodtger U, *et al.* Transthoracic ultrasound-guided biopsy in the hands of chest physicians – a stepwise approach. *Eur Clin Respir J* 2019; 6: 1579632.
7 Yamamoto N, Watanabe T, Yamada K, *et al.* Efficacy and safety of ultrasound (US) guided percutaneous needle biopsy for peripheral lung or pleural lesion: comparison with computed tomography (CT) guided needle biopsy. *J Thorac Dis* 2019; 11: 936–943.
8 Zhang H, Guang Y, He W, *et al.* Ultrasound-guided percutaneous needle biopsy skill for peripheral lung lesions and complications prevention. *J Thorac Dis* 2020; 12: 3697–3705.
9 Laursen CB, Clive A, Hallifax R, *et al.* European Respiratory Society statement on thoracic ultrasound. *Eur Respir J* 2021; 57: 2001519.
10 Lemieux S, Kim T, Pothier-Piccinin O, *et al.* Ultrasound-guided transthoracic needle biopsy of the lung: sensitivity and safety variables. *Eur Radiol* 2021; 31: 8272–8281.
11 Sconfienza LM, Mauri G, Grossi F, *et al.* Pleural and peripheral lung lesions: comparison of US- and CT-guided biopsy. *Radiology* 2013; 266: 930–935.
12 Hsu WH, Ikezoe J, Chen CY, *et al.* Color Doppler ultrasound signals of thoracic lesions. Correlation with resected histologic specimens. *Am J Respir Crit Care Med* 1996; 153: 1938–1951.
13 Laursen CB, Graumann O, Moller TV, *et al.* Contrast-enhanced ultrasound-guided transthoracic lung biopsy. *Am J Respir Crit Care Med* 2016; 194: e5–e6.
14 Kuo YW, Chen YL, Wu HD, *et al.* Application of transthoracic shear-wave ultrasound elastography in lung lesions. *Eur Respir J* 2021; 57: 2002347.
15 Lim CK, Chung CL, Lin YT, *et al.* Transthoracic ultrasound elastography in pulmonary lesions and diseases. *Ultrasound Med Biol* 2017; 43: 145–152.
16 Gorguner M, Misirlioglu F, Polat P, *et al.* Color Doppler sonographically guided transthoracic needle aspiration of lung and mediastinal masses. *J Ultrasound Med* 2003; 22: 703–708.
17 Salamonsen M, Dobeli K, McGrath D, *et al.* Physician-performed ultrasound can accurately screen for a vulnerable intercostal artery prior to chest drainage procedures. *Respirology* 2013; 18: 942–947.
18 Liao WY, Jerng JS, Chen KY, *et al.* Value of imprint cytology for ultrasound-guided transthoracic core biopsy. *Eur Respir J* 2004; 24: 905–909.
19 Jiang B, Li XL, Yin Y, *et al.* Ultrasound elastography: a novel tool for the differential diagnosis of pleural effusion. *Eur Respir J* 2019; 54: 1802018.
20 Chen YL, Kuo YW, Wu HD, *et al.* The application of ultrasound shear wave elastography in the prediction of paradoxical upgrading reaction in tuberculous lymphadenitis. A pilot study. *J Formos Med Assoc* 2022; 121: 1696–1704.
21 Takayama N, Ishiguro Y, Taniguchi N, *et al.* The effect of ultrasound with acoustic radiation force on rabbit lung tissue: a preliminary study. *J Med Ultrason* 2016; 43: 481–485.
22 Miller DL, Dong Z, Dou C, *et al.* Pulmonary capillary hemorrhage induced by acoustic radiation force impulse shear wave elastography in ventilated rats. *J Ultrasound Med* 2019; 38: 2575–2587.

23 Hartman C, Child SZ, Mayer R, *et al.* Lung damage from exposure to the fields of an electrohydraulic lithotripter. *Ultrasound Med Biol* 1990; 16: 675–679.
24 Gonzalez AV, Silvestri GA, Korevaar DA, *et al.* Assessment of advanced diagnostic bronchoscopy outcomes for peripheral lung lesions: a Delphi consensus definition of diagnostic yield and recommendations for patient-centered study designs. An official American Thoracic Society/American College of Chest Physicians research statement. *Am J Respir Crit Care Med* 2024; 209: 634–646.
25 Gennisson JL, Deffieux T, Fink M, *et al.* Ultrasound elastography: principles and techniques. *Diagn Interv Imaging* 2013; 94: 487–495.
26 Safai Zadeh E, Gorg C, Prosch H, *et al.* WFUMB technological review: how to perform contrast-enhanced ultrasound of the lung. *Ultrasound Med Biol* 2022; 48: 598–616.
27 Jacobsen N, Pietersen PI, Nolsoe C, *et al.* Clinical applications of contrast-enhanced thoracic ultrasound (CETUS) compared to standard reference tests: a systematic review. *Ultraschall Med* 2022; 43: 72–81.
28 Fergusson D, Aaron SD, Guyatt G, *et al.* Post-randomisation exclusions: the intention to treat principle and excluding patients from analysis. *BMJ* 2002; 325: 652–654.